CS & IT ENGINEERING



Operating System

Disk Scheduling Algorithm

DDP_01 Discussion Notes



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[NAT]

#Q. A particular disk unit uses a bit string to record the occupancy or vacancy of its disk blocks with '0' denoting occupied block and '1' denoting vacant block. A 32-bit part of this string has Hexadecimal value of AC0121DE. The percentage of occupied blocks on the disk for this part is 59.4 % (rounded to 1 decimal place)?

$(AC0121DE)_{16} = (1010\ 1100\ 0000\ 0001\ 0010\ 0001\ 1101\ 1110)_2$

↳ 19 → 0's

% occupied block $= \frac{19}{32} = 59.375\ \% \simeq 59.4\ \%$

Ans 59.4

[NAT]

#Q. A system directory is kept in 2 disk blocks each of size 4Kbytes. It is a single level-directory and each directory entry is of size 64-bits. The maximum number of files possible in this system is __1__ k?

$1k = 2^{10}$

System directory occupied space = $2 \times 4kB = 8kB$

Each directory entry size = 64 bits = 8B.

Max number of entries possible in file system = $\frac{8kB}{8B}$

$= 1k.$

Ans 1

[NAT]

#Q. The index node (inode) of a unix-like file system has 10 direct, one single-indirect, one double-indirect and one triple-indirect pointer. The disk block size is 2KB and disk block addresses 64-bits. The maximum possible file size (rounded off to nearest integer) is 32 GB?

Disk block size = 2KB.

Disk block address = 64 bits = 8B.

$1GB = 2^{30}B$

number address per block = $\frac{2KB}{8B} = 2^8$

The file size = $10 \times 2KB + 2^8 \times 2KB + 2^8 \times 2^8 \times 2KB + 2^8 \times 2^8 \times 2^8 \times 2KB$.

$= \frac{10}{2^{19}} GB + \frac{1}{2^{11}} GB + \frac{1}{2^3} GB + 32\ GB.$

$= 32.125\ GB.$

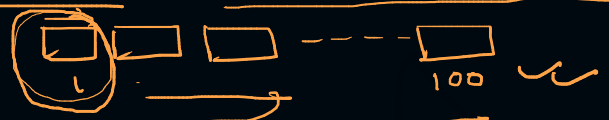
$= 32\ GB.$

Ans 32

[MSQ]

#Q. Consider a file which is stored on disk with either contiguous file allocation method or linked file allocation method or indexed file allocation method. The file is stored on 100 blocks. Which of the following will take minimum number of disk block accesses of the file for insertion of a new block?

A. Insertion of a new block at starting if file is stored using contiguous file allocation method. 100 block access.



B. Insertion of a new block after 50th block of file if file is stored using linked file allocation method. 50 block access.



C. Insertion of a new block in the end if file is stored using indexed file allocation method. 1 block access.

D. Insertion of a new block in the end if file is stored using contiguous file allocation method. 1 block access.

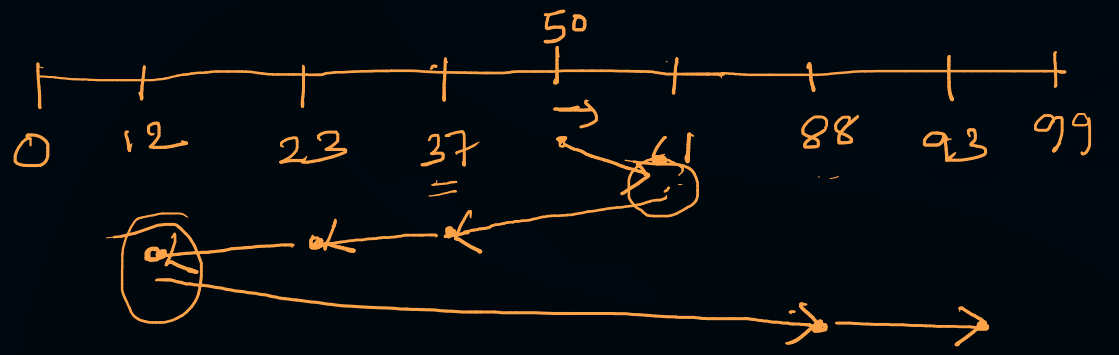
Ans C, D.

[NAT]

#Q. Consider a disk with 6 platters each with 2 recording surfaces and 100 cylinders (numbered as 0, 1, 2 ...., 99). The following 6 disk requests (for cylinder number) are received by disk controller at the same time:

61, 88, 23, 37, 12, 93

Currently the head is positioned at cylinder number 50 and is moving towards higher cylinder numbers. The average seek time in moving head over 3 cylinder is 2 milliseconds and for reversing the direction of the head once is 5 millisecond. The total seek time to satisfy all the above requests using the Shortest seek time first disk scheduling algorithm is 104 ?



Ans 104.

Total head movement = (61-50) + (61-12) + (93-12)

= 11 + 130

= 141

3 cylinder → 2ms.

1 → $\frac{2}{3}$

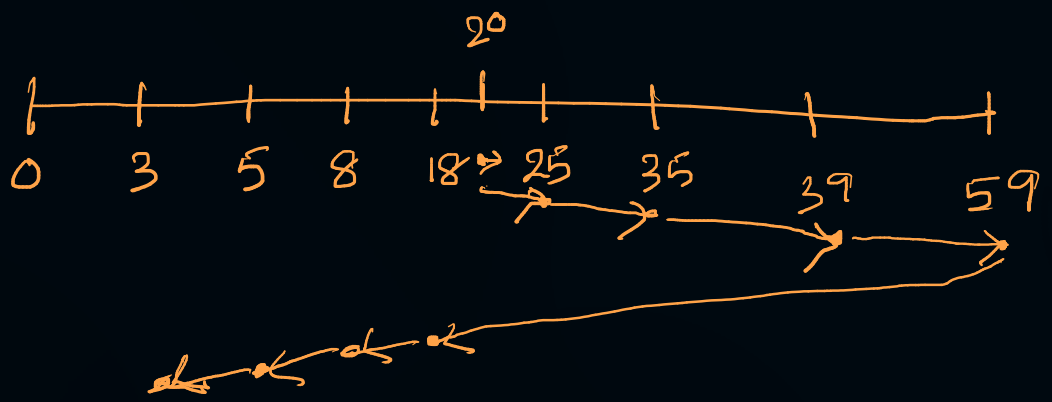
141 → $\frac{2}{3} \times 141^{47}$

= 94 ms

Total seek time will be 94 + 5×2 = 104 ms

[NAT]

#Q. Consider a disk with 60 cylinders (numbered 0, 1, 2 ....59). Disk requests are made for a disk drive for cylinders 5, 25, 18, 3, 39, 8 and 35 in that order. Assume that the arm is current at cylinder 20 and moving towards higher cylinder numbers. The above all requests are serviced using scan algorithm. The number of head movements are 95. ?



Total head movements = (59 − 20) + (59 − 3)

= 39 + 56 = 95

Ans 95

THANK - YOU